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Taxonomic assignment of the Cambrian trilobite *Tonkinella* Mansuy, 1916 (Corynexochida), with a new species from California

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Tonkinella has been assigned to different families and subfamilies within the Oryctocephalidae, including the Tonkinellidae and Tonkinellinae, with other genera and by itself. *Tonkinella* is hypothesised to have either evolved from the spinose oryctocephalids (Oryctocephalinae) or from the nonspinose oryctocephalids (Oryctocarinae). Phylogenetic analysis indicates that the *Tonkinella* clade and the *Arthricocephalus* clade are not immediate sister groups within the spinose oryctocephalid clade; they are separated by *Metabalangia*. However, the very long branch length leading to the *Arthricocephalus* clade suggests that genus and its close relatives do not belong within the spinose oryctocephalids. These results confirm earlier suggestions that *Tonkinella* is probably derived from the spinose oryctocephalids and should be included in the Oryctocephalinae. The phylogenetic analysis also shows that: *Barklyella expansa* falls within the *Tonkinella* clade; *Oryctocephaloides* is polymorphic; and that the *Cheiruroides* and *Arthricocephalus* clades are not sister groups. Furthermore, *Tonkinella spinosa* n. sp., which has pygidial spines, could have been a link between the spinose oryctocephalids and other species of *Tonkinella*. However, the position of the species within the cladogram indicates that the pygidial spines resulted from character reversal representing a homeotic mutation.

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TONKINELLA Mansuy, 1916, is a widely distributed corynexochid trilobite genus found in North America (Resser 1938b; Rasetti 1951; Young & Ludvigsen 1989; Sundberg 1994; Stewart & Poole 2002), Argentina (Bordonaro & Baldis 1987; Bordonaro & Banchig 1995), India (Reed 1934; Jell & Hughes 1997), China (Zhang 1957, 1959, 1998; Yuan *et al.* 2001a; Yuan & Li 2004), Korea (Kobayashi 1962; Hong *et al.* 2003), Vietnam (Mansuy 1916), Russia (Chernysheva 1952; Repina 1960; Egorova & Savitskiy 1969; Savitskiy *et al.* 1972; Dalmatov 1975), and Spain (Liñán & Gozalo 1999; Gozalo *et al.* 2003). *Tonkinella* species are commonly found in the *Ptychagnostus gibbus* Biozone; however, they occur earlier in the Lower Cambrian of Spain with *T. sequei* Gozalo (in Liñán & Gozalo 1999) and in the *Kounamkites* Biozone (lower 'Middle' Cambrian) of Siberia with *T. gavrilovae* Dalmatov, 1975 and *T. sisimica* Repina, 1960 (=*T. breviceps* Kobayashi, 1934) (Fig. 1). Due to limited biostratigraphic range, widespread distribution of the genus, and distinctive cranidia and pygidia, *Tonkinella* has been used as a guide fossil for the 'Middle' Cambrian (Cambrian Series 3) around the world (e.g., Gozalo *et al.* 2003; Sundberg 2005).

Tonkinella spinosa n. sp. is the only known species of *Tonkinella* that possesses pygidial spines. The occurrence of *Tonkinella spinosa* in the Emigrant Formation is lower than most *Tonkinella*, and it probably represents the *Albertella* or *Glossopleura* biochron (see 'stratigraphy' below). The low stratigraphic position and the pygidial spines suggest that *T. spinosa* is intermediate between older spinose oryctocephalids (e.g., *Oryctocephalus* Walcott, 1886 or *Oryctocephalites* Resser, 1939) and more typical *Tonkinella*. This further suggests that *Tonkinella* is more closely related to the spinose Oryctocephalinae Beecher, 1897 rather than the nonspinose Oryctocarinae Hupé, 1953 as suggested by some authors. A phylogenetic analysis of the different species of *Tonkinella* and representatives of both spinose and nonspinose oryctocephalids is presented here to test this suggested relationship.

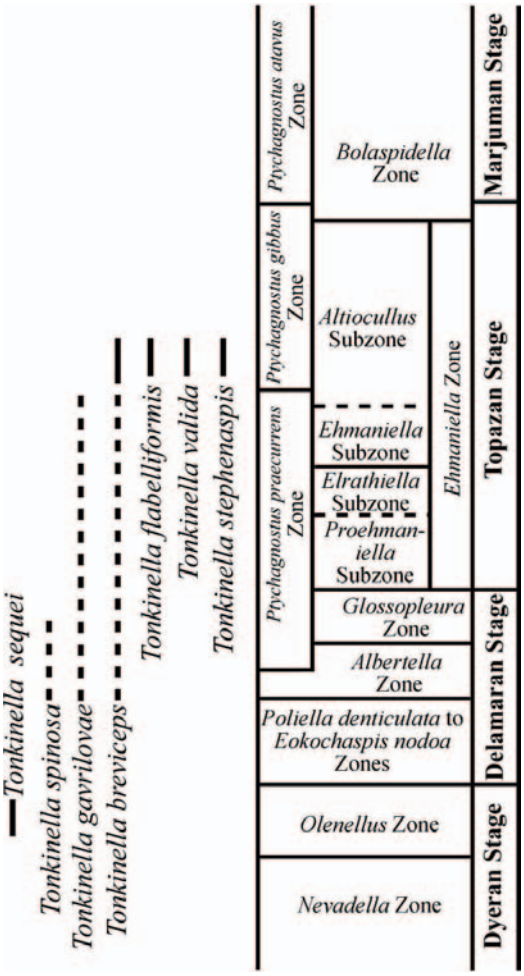

Fig. 1. Stratigraphic ranges of species of *Tonkinella* assigned to the Laurentian stages (see Palmer 1998; Sundberg 2005). The dashed line for *T. spinosa* represents the approximate levels that the species is thought to range. The stratigraphic range of *T. sibirica* is not known besides being 'Middle' Cambrian and the dashed lines for *T. breviceps* and *T. gavrilovae* represent the possible range of the *Kounamkites* Biozone of Siberia (T. Pegel, pers. comm., 2005).

PREVIOUS TAXONOMIC ASSIGNMENT OF *TONKINELLA*

The family Oryctocephalidae Beecher, 1897 was subdivided into two subfamilies, the spinose Oryctocephalinae and the nonspinose Oryctocarinae by Hupé (1953, also see Whittington 1995). Kobayashi (1961) subdivided the family into three subfamilies (Lancastriinae Kobayashi, 1935, Oryctocephalinae, and Tonkinellinae Hupé, 1953 [=Oryctocarinae]). Suvorova (1964) subdivided the family into five subfamilies (Oryctocephalinae, Oryctocarinae, Tonkinellinae, Cheiruroidinae Suvorova, 1964, and Lancastriinae). Oryctocephalids were also referred by Zhang *et al.* (1980) to the superfamily Oryctocephalacea Beecher, 1897, which they subdivided into five families and six subfamilies (Oryctocephalidae with Lancastriinae, Protoryctocephalinae Qian in Zhang *et al.*, 1980, and Oryctocephalinae; Feilongshaniidae Qian & Lin in Zhang *et al.*, 1980 with Feilongshaniinae Qian & Lin in Zhang *et al.*, 1980, Oryctocarinae Hupé, 1953, and Sandoverinae Qian in Zhang *et al.*, 1980; Cheiruroideidae Zhang, 1963; Tonkinellidae Hupé, 1953; and Balangiidae Chien, 1961). Recently, Yuan *et al.* (2002) defined a new subfamily, Curvoryctocephalinae Zhao & Yuan, within the family Oryctocephalidae. Only the spinose oryctocephalids have been analysed using cladistic methods (Sundberg & McCollum 1997).

The placement of *Tonkinella* within the oryctocephalid hierarchy has varied. Reed (1934, p. 9) erected the family Tonkinellidae to embrace *Tonkinella*, although he provided no diagnosis or comment on the new family, and did not even note that the family was new. Hupé (1953) questionably placed *Tonkinella* into a subfamily, Tonkinellinae, within the family Oryctocephalidae. Kobayashi (1961) placed the genus into the subfamily Tonkinellinae with *Oryctocara* Walcott, 1908, *Arthricocephalus* Bergeron, 1899, and ?*Cheiruroides* Kobayashi, 1935. Zhang *et al.* (1980, p. 425) placed the genus in the monogeneric family Tonkinellidae. Whittington (1995) placed the genus into the Oryctocephalinae with the spinose oryctocephalids. Yuan *et al.* (2001b) placed *Tonkinella* in the Tonkinellinae with five other genera. McNamara *et al.* (2006) suggested a relationship between *Metabalangia* Qian & Yuan (in Zhang *et al.*, 1980) and *Tonkinella* and recognised the presence of three subfamilies, Oryctocephalinae, Tonkinellinae and Oryctocarinae.

Zhang (1980), McNamara (1986), Whittington (1995), Yuan *et al.* (2001b), and McNamara *et al.* (2006) are the only authors that previously hypothesised relationships of *Tonkinella* to other oryctocephalid groups. Whittington (1995) maintained only the two subfamilies previously recognised by Hupé (1953) and placed *Tonkinella* in the Oryctocephalinae based on the presence of a long genal spine and the suggestion (McNamara 1986) that it evolved from spinose oryctocephalids (possibly *Oryctocephalus* [= *Oryctocephalites* as defined by Sundberg & McCollum 1997]) through paedomorphosis. McNamara *et al.* (2006) later suggested a close relationship between the spinose oryctocephalid *Metabalangia* and *Tonkinella*. Zhang (1980) suggested that *Tonkinella* was derived from the

subfamily Oryctocarinae. Yuan *et al.* (2001b) proposed that the subfamily Tonkinellinae contained the genera *Arthricocephalus*, *Arthricocephalites* Chien & Lin in Lu *et al.*, 1974 (=*Arthricocephalus* see McNamara *et al.* 2003), *Ovatoryctocara* Chernysheva, 1962, *Udjanella* Solovyev, 1964, *Barklyella* Shergold, 1969, and *Tonkinella*. The subfamily was placed in the family Oryctocaridae, which they thought was derived from *Cheiruroides*. Within the subfamily Tonkinellinae, Yuan *et al.* (2001b, text-fig. 1) recognised two lineages branching from '*Arthricocephalites*'. The first lineage includes *Ovatoryctocara* to *Udjanella*, the second lineage includes *Barklyella* to *Tonkinella*. For the latter lineage, they cited the progressive reduction in thoracic segments, reduction in the pygidial axis length, effacement of pygidial interpleural furrows and border, and changes from proparian to gonatoparian to opisthoparian sutures. None of these relationships have been tested.

PHYLOGENETIC ANALYSIS

Methods

Twenty-four oryctocephalid trilobite species were analysed phylogenetically. *Tonkinella* species include most of the described taxa: *T. breviceps*; *T. flabelliformis* Mansuy, 1916 (type species); *T.* ex. gr. *flabelliformis* of Chernysheva, 1952; *T. gavrilovae*; *T. sequei*; *T. sibirica* Chernysheva, 1952; *T. spinosa*; *T. stephensis* Kobayashi, 1935; and *T. valida* Chernysheva, 1962. *Tonkinella appalachia* Resser, 1938b; *T. kobayashi* Resser, 1938c; *T. sisimica*; *T. khorbusuonkensis* Solovyev, 1989; and *T. occidentalis* Young & Ludvigsen, 1989 (partim) are junior synonyms of *T. breviceps* (see Sundberg 1994). *Tonkinella occidentalis* (partim) is also a junior synonym of *T. valida* (see Sundberg 1994). *Tonkinella shantungensis* Zhang, 1959 is a junior synonym of *T. sibirica*. *Tonkinella bambuica* Dalmatov, 1975 is known from a single distorted, but complete specimen. This species was not included in this analysis because of difficulty assessing the morphology of the single known specimen. *Barklyella expansa* Shergold, 1969 was also included because it was originally aligned with *Tonkinella* by Shergold (1969) and was thought to be the ancestor of *Tonkinella* by Yuan *et al.* (2001b). Other taxa in the study were included because they either represent the most plesiomorphic representatives of some groups (see Sundberg & McCollum 1997), because of morphological similarity, or because of suggested phylogenetic relationships (McNamara 1986; Yuan *et al.* 2001b). *Oryctocephalus lancastrioides* (Shergold, 1969), *Lancastria roddyi* (Walcott, 1912) and *Changaspis elongata* Lee (in Chien, 1961; McNamara *et al.* 2006 considered *C. placenta* Qian & Lin [in Zhang *et al.,* 1980] and *C. micropyge* Chien, 1961 as junior synonyms of *C. elongata*) represent the most plesiomorphic representatives of the spinose oryctocephalids based on a cladistic analysis (Sundberg & McCollum 1997). McNamara (1986) suggested that *Tonkinella* was derived from *Oryctocephalites*. *Oryctocephalites rasettii* Sundberg & McCollum, 1997 and *O. palmeri* Sundberg & McCollum, 1997 represent some of the most plesiomorphic representatives of the genus (see Sundberg & McCollum 1997). *Oryctocephalites typicalis* Resser, 1939, *Oryctocephaloides convexus* Yuan (in Zhang *et al.*, 1980), and *Oryctocephaloides robustus* Zhao & Yuan (in Yuan *et al.*, 2001b) were selected due to their overall similarity to *Tonkinella*. McNamara *et al.* (2006) suggested that *Metabalangia* should also be assigned to Tonkinellinae; thus, the type species *M. yupingensis* Qian & Yuan (in Zhang *et al.,* 1980) was included in these analyses. *Metabalangia transversa* Zhao & Yuan (in Yuan *et al.*, 2002) is considered here a junior synonym of the type species because it is found in the same stratigraphic horizons, at the same localities, and has only minor morphological differences (i.e., gentler arched anterior margin of the cranidium, slightly broader glabella, slightly narrower fixigena, five thoracic segments and slightly longer stout genal spine), most of which can be the result of the slight compression of the shale specimens. *Arthricocephalus chauveaui* Bergeron, 1899, '*Arthricocephalites*' *xinzhaiheensis* Chien & Lin (in Lu *et al.*, 1974), *Cheiruroides primigenius* (Saito, 1934) and *C. arcticus* Chernysheva, 1962 were selected based on the proposed phylogeny of Yuan *et al.* (2001b).

Redlichia is used as the outgroup based on a preliminary cladistic analysis of Cambrian Series 2-3 (lower and middle Cambrian) trilobite hypostomes. Hypostomes appear to be one of the most conservative features of trilobites. The spinose oryctocephalids (Shergold 1969; Blaker & Peel 1997; Sundberg & McCollum 1997) and *Tonkinella* (Jell & Hughes 1997) have a fused hypostome and rostral plate, shallow macula, an oval anterior lobe, a posterior lobe that has sides nearly reaching the middle of the anterior lobe, a well developed lateral border that nearly reaches the rostral plate, and weak or absent posterior and anterior wings. *Redlichia* has a hypostome most similar to the oryctocephalids (see Zhang & Jell 1987, pl. 7, figs 2, 8; Kruse 1998, fig. 11H). The cranidial, thoracic, and pygidial character states for *Redlichia* for this phylogenetic study of *Tonkinella* are based on the species illustrated by Zhang *et al.* (1980), Zhang & Jell (1987) and Kruse (1998). The states were assigned to the

closest state present in the oryctocephalids (e.g., the palpebral lobe length of *Redlichia* is much greater than any oryctocephalid, thus the state was assigned to the longest lobe length present in the oryctocephalids).

Twenty characters were compiled for the cladistic analyses (Appendix 1). Methods used are the same as those outlined by Sundberg (2004). A base weight of 1000 steps was used. The division of continuous characters into individual states is based on a modified segment coding method as described by Sundberg (2004), which is similar to that used by Chappill (1989). Justification for and the importance of the use of continuous characters have been discussed by Sundberg (1999, 2004) and Poe & Wiens (2000).

The *Tonkinella* classification was based on several morphological features. Characteristics that have been commonly used include: direction of ocular ridges; upturned anterior cranidial border; convexity of glabellar frontal lobe; presence of a plectrum; presence of a medial sulcus; smoothness of the pygidial margin; depth of pygidial pleural furrows; pygidial pleural furrows reaching border; width of the posterior pleural rib of the pygidium; and number of pygidial axial rings. Characters generally accepted as distinctive of *Tonkinella* include: lateral glabellar furrows connecting with axial furrows at S2 and S3; absence of pygidial interpleural furrows; absence of pygidial spines; and the presence of genal spines. Additional characters used to define the subfamily or family are: an expanding glabella; opisthoparian suture; size of the pygidium; length of the pygidial axis; and effacement of pygidial interpleural furrows and border. Most of these characters (see Appendix 1) were used in the phylogenetic analysis to determine the relationships between the different species of *Tonkinella* and their relationship to other oryctocephalids. Additional characters used in the analysis include: the number of transglabellar furrows; the shape of lateral glabellar furrows; glabellar shape; length of palpebral lobes; presence of a posterior lobe of the hypostome; a non-triangular termination of the rostral plate; number of thoracic segments; and presence of pleural spines on the thorax. These characters were considered important in identifying other groups of oryctocephalids (e.g., Suvorova 1964; Shergold 1969; Sundberg 1994; Whittington 1995; Sundberg & McCollum 1997; Yuan *et al.* 2001b, 2002; and the preliminary hypostome analysis).

Ontogenetic character changes were not directly used in the analyses. Although these characters could potentially provide information of phylogenetic relationships, the present state of knowledge of oryctocephalid ontogeny is limited. The ontogeny of only a few Oryctocarinae is known (Suvorova 1964; McNamara *et al.* 2003, 2006). Only two Oryctocephalinae ontogenies are known, one only partially (Suvorova 1964; McNamara *et al.* 2006), and no ontogenetic information is known about *Tonkinella*. However, characters that that change in ontogeny were included in the analyses (e.g., characters 2, 3, 4, 5, 6, 12, 14, 17, and 18).

Cladograms were generated using PAUP's heuristic search with random addition of taxa (1000 repetitions) and monophyletic ingroup rooting (Swofford 2003). ACCTRAN optimization was employed. In the two analyses of *Tonkinella* and the other oryctocephalids, character states were considered ordered except for characters 2 to 5. Appendix 2 lists the character states for each species. A third analysis of just the *Tonkinella* species was also conducted. The outgroup chosen for this analysis was *Metabalangia yupingensis*. This species was discovered to be the sister group to *Tonkinella* in the first two analyses described below. Character states were considered ordered. Appendices 4 and 5 list the characters and character states for each species.

Results

Three different cladistic analyses were conducted. The first, which included all taxa, tested Yuan *et al.*'s (2001b), McNamara's (1986) and McNamara *et al.*'s (2006) phylogenetic hypotheses. In the second analysis, which is justified below in the discussion, *Cheiruroides*, *Arthricocephalus*, and '*Arthricocephalites*' were eliminated from the analysis. This analysis further tested McNamara's (1986) and McNamara *et al.*'s (2006) phylogenetic hypotheses. The third analysis focused on the relationships within *Tonkinella* and used species-defining characters.

The first analysis yielded 30 most-parsimonious cladograms with a tree length of 21,392 steps, consistency index of 0.7359 (excluding uninformative characters), homoplasy index of 0.2641 (excluding uninformative characters), rescaled consistency index of 0.6785, and retention index of 0.8922 (Fig. 2). The consistency index is higher than expected given the number of taxa analysed and that the study contained several ordered characters (see Sanderson & Donoghue 1989 for comparative values). The apomorphic changes at the nodes are listed in Appendix 3. Four major clades were discovered, the *Lancastria* clade (*L. roddyi*, *Oryctocephalus lancastrioides* and *Changaspis placenta*), the *Cheiruroides* clade (*Cheiruroides primigenius* and *C. arcticus*), the *Arthricocephalus* clade (*Arthricocephalus chauveaui* and '*Arthricocephalites*' *xinzhaiheensis*), and the

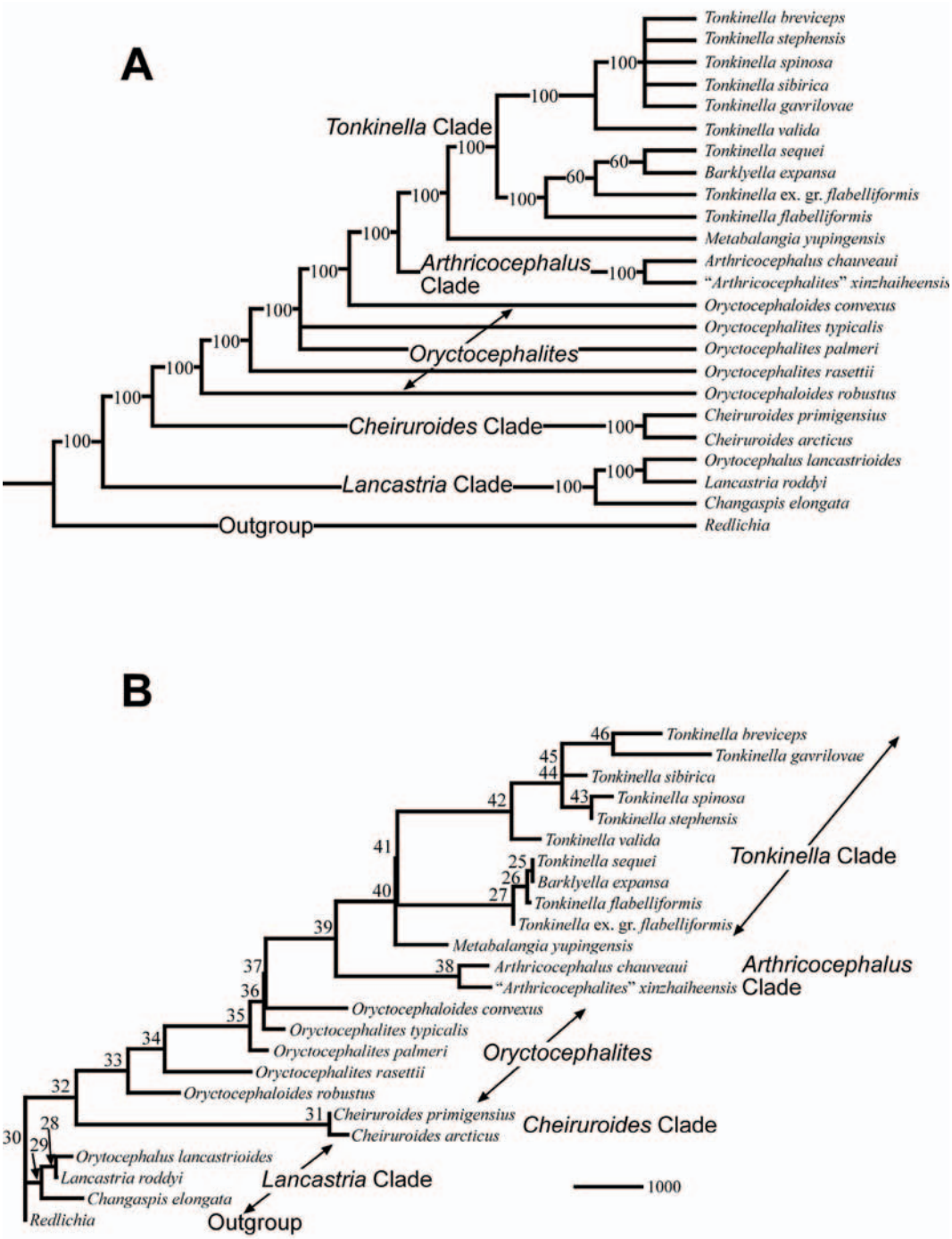

Fig. 2. The majority rule consensus cladogram (A) and the first phylogram of 30 phylograms (B) of the representatives of Oryctocephalinae, *Tonkinella* and 'Oryctocarinae' (*Arthricocephalus* and *Cheiruroides* clades). See Appendix 3 for character changes at the different nodes.

Tonkinella clade (including all *Tonkinella* species). The *Oryctocephalites-Oryctocephaloides-Metabalangia* taxa have a stepwise distribution between the *Lancastria* and the *Tonkinella* clades. The *Arthricocephalus* and *Tonkinella* clades are sister groups with *Metabalangia yupingensis* occurring between the two clades. *Barklyella* falls in the *Tonkinella* clade.

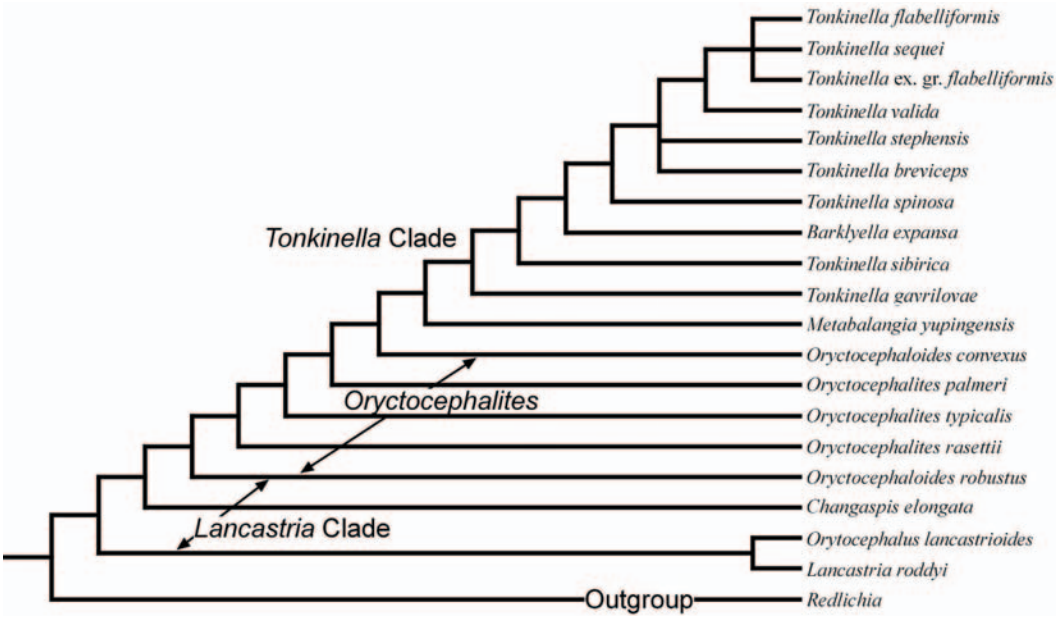

Fig. 3. The strict consensus cladogram of the representatives of Oryctocephalinae and *Tonkinella*.

The *Lancastria* clade is defined at node 29 by the absence of glabellar furrows connected to the axial furrows [3(0)], pitted lateral glabellar furrows [4(0)], relatively long palpebral lobes [6(2)], posterolaterally directed ocular ridges [7(2)], and moderate length pygidial axis [18(2)]. The *Cheiruroides* clade is defined at node 31 by gonatoparian facial sutures [1(1)], relatively short palpebral lobes [6(0)], the absence of genal spines [9(0)], the absence of the posterior lobe of the hypostome [10(1)], the presence of a triangular termination of the rostral plate [11(1)], the absence of pleural spines [13(0)], and the absence of pygidial spines [15(0)]. The combined *Tonkinella-Arthricocephalus* clades are defined at node 39 by the absence of pygidial spines [15(0)], four axial rings in the pygidium [17(4)], and the presence of a pygidial border [20(1)]. The *Arthricocephalus* clade is defined at node 38 by proparian facial sutures [1(1)], three transglabellar furrows [2(3)], anteriorly expanded glabella [5(2)], absence of genal spines [9(0)], absence of pleural spines [13(0)], and moderate length of the pygidial axis [18(2)]. The *Tonkinella* clade is defined at node 41 by three glabellar furrows connected to the axial furrows [3(3)], anteriorly expanded glabella [5(2)], absence of pygidial interpleural furrows [16(0)], and moderate pygidial convexity [19(1)].

The second analysis excludes the *Cheiruroides* and *Arthricocephalus* clades. The analysis yielded six most parsimonious cladograms with a tree length of 19,510 steps, consistency index of 0.7530 (excluding uninformative characters), homoplasy index of 0.2470 (excluding uninformative characters), rescaled consistency index of 0.6923, and retention index of 0.9042 (Fig. 3).

This second analysis was essentially the same as the one using all taxa, except that the *Cheiruroides* and *Arthricocephalus* clades were excluded. In the *Lancastria* clade, *Changaspis elongata* appears at the base of the *Oryctocephalites-Tonkinella* clade (see also Sundberg & McCollum 1997). The *Tonkinella* clade is the most derived group of the spinose oryctocephalids; however, the order of the species in the clade is not the same as in the first analysis. The *Tonkinella* clade is defined by one pair of lateral glabellar furrows connected to the axial furrows [3(1)], anteriorly expanded glabella [5(2)], six pygidial axial rings [17(6)], and moderate pygidial convexity [19(1)]. The difference in some of the node character states from the first analysis is the result of the rearrangement of species within the *Tonkinella* clade.

The third analysis was done to help establish the phylogenetic relationships of the different *Tonkinella* species. This analysis yielded one most-parsimonious cladogram with a tree length of 20,169 steps, consistency index of 0.9372, homoplasy index of 0.2245, rescaled consistency index of 0.9094, and retention index of 0.8524 (Fig. 4). The consistency index is close to the expected value given the number of taxa analyzed (see Sanderson & Donoghue 1989 for comparative values). The *Tonkinella* clade is defined at node 13 and by the following character states: moderately deep S1 transglabellar furrow [3(2)], a shallow S2

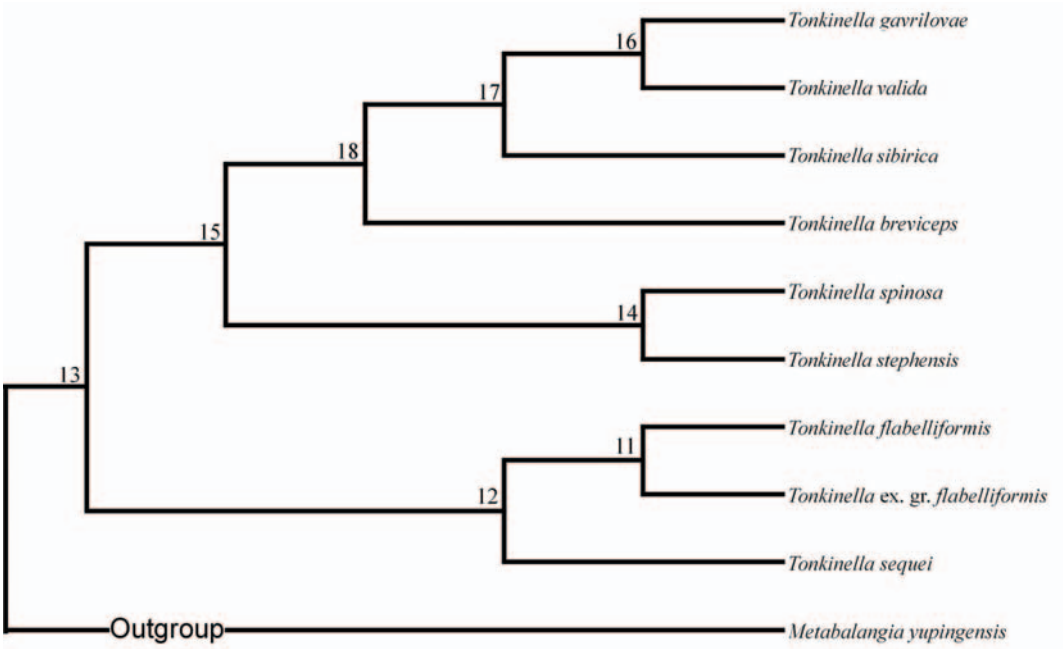

Fig. 4. The single most parsimonious cladogram of the representatives of *Metabalangia* and *Tonkinella*. See appendix 6 for character changes at the different nodes.

transglabellar furrow [4(1)], moderately directed ocular ridge [7(2)], and shallow pleural furrows on the pygidium [13(1)] (Appendix 6).

Tonkinella is divided into two major clades, the *T. flabelliformis* and *T. breviceps* clades. The *T. flabelliformis* clade contains *T. flabelliformis*, *T.* ex. gr. *flabelliformis*, and *T. sequei* and is defined at node 12 by the shallow depth of the S3 transglabellar furrow [5(1)], weak ocular ridge [6(1)], pleural furrows that do not reach the pygidial margin [14(0)], and moderately short length of the pygidial axis [18(1)]. The *T. breviceps* clade is defined at node 15 by moderate expansion of the frontal lobe [1(1)], no curvature of the anterior border of the pygidium [12(0)], 13 pleural ribs [15(3)], and five axial rings of the pygidium [19(5)]. This clade contains two subclades: the first clade contains *T. breviceps*, *T. sibirica*, *T. valida*, and *T. gavrilovae*; and the second clade contains *T. spinosa* and *T. stephensis*. The first clade is defined at node 18 by a shallow S3 transglabellar furrow [5(1)], weak ocular ridge [6(1)], and the pleural furrows that do not reach the pygidial margin [14(0)]. The second clade is defined at node 14 by a scalloped pygidial margin [11(1)].

DISCUSSION

Yuan *et al.*'s (2001b) hypotheses that *Tonkinella* evolved from the lineage of *Arthricocephalus* to '*Arthricocephalites*' to *Barklyella* and that *Arthricocephalus* evolved from *Cheiruroides* are not supported by the cladistic analysis. The *Arthricocephalus* clade is a sister clade to the *Tonkinella-Metabalangia* clade and is removed from the *Cheiruroides* clade. *Barklyella* occurs within the *Tonkinella* clade. Zhang's (1980) suggestion that *Tonkinella* evolved from Oryctocarinae is also not supported. McNamara's (1986) hypothesis that *Tonkinella* evolved from the spinose oryctocephalids is supported by the cladogram illustrating a close relationship between *Oryctocephalites* and *Tonkinella*. McNamara *et al.*'s (2006) hypothesis that *Metabalangia* is closely related to *Tonkinella* is strongly supported in the cladogram given that *M. yupingensis* is the sister group to *Tonkinella*. The branch lengths separating *Oryctocephalites* from *Metabalangia* and *Tonkinella* are relatively short, suggesting that *Tonkinella* was derived from *Oryctocephalites* with *Oryctocephaloides convexus* and *Metabalangia* falling inbetween the two groups.

One of the difficulties in establishing a phylogenetic relationship of *Tonkinella* is that the cladistics forces a solution on a data set. There are three pygidial characters linking the *Tonkinella-Metabalangia* clade with the *Arthricocephalus* clade, the absence of pygidial spines [15(0)], four pygidial axial segments [17(4)], and the presence of a pygidial border [20(1)]. However, the branch length separating the *Arthricocephalus* clade from the remaining portion is quite long, possibly indicating that the *Arthricocephalus* clade is

Character	Type	Original Weight	Final Weight	Percent of original weight
1	Ord	1000	444	44
2	Unord	1000	200	20
3	--	1000	133	13
4	--	1000	48	5
5	Unord	1000	63	6
6	Ord	333	6	2
7	Ord	333	13	4
8	--	1000	200	20
9	--	1000	0	0
10	--	1000	1000	100
11	--	1000	1000	100
12	Ord	1000	515	52
13	--	1000	167	17
14	Ord	1000	1000	100
15	--	1000	259	26
16	--	1000	429	43
17	Ord	1000	195	20
18	--	250	25	10
19	--	1000	1000	100
20	--	1000	400	40

Table 1. Percentage of change in character weights resulting from progressive reweighting of characters based on RC of the cladograms from the first analysis including the *Arthricocephalus* and *Cheiruroides* clades. Ord = ordered character states, Unord = unordered character states.

Character	Type	Status	Original Weight	Final Weight	Percent of original weight
1	--	UC	1000	1000	100
2	Unord	U	1000	0	0
3	--	-	1000	333	33
4	--	-	1000	28	3
5	Unord	-	1000	333	33
6	Ord	-	333	0	0
7	Ord	-	333	60	18
8	--	-	1000	200	20
9	--	UC	1000	1000	100
10	--	UC	1000	1000	100
11	--	UC	1000	1000	100
12	Ord	-	1000	604	60
13	--	U	1000	1000	100
14	Ord	-	1000	1000	100
15	--	-	1000	438	44
16	--	-	1000	1000	100
17	Ord	-	1000	173	17
18	--	-	250	44	18
19	--	-	1000	1000	100
20	--	-	1000	333	33

Table 2. Percentage of change in character weights resulting from progressive reweighting of characters based on RC of the *Tonkinella* and spiny oryctocephalid cladogram. Ord = ordered character states, Unord = unordered character states, U = uninformative, UC = uninformative, constant.

not derived from the spinose oryctocephalids as suggested by the cladogram. A long branch length in the *Cheiruroides* clade also suggests that the genus does not belong in the oryctocephalid clade. Supporting this is the unusual hypostome of *Cheiruroides arcticus*, which is unlike any spinose oryctocephalid hypostome.

Based on the presence of pygidial spines and its relatively older age, *Tonkinella spinosa* could be interpreted as an intermediate between *Oryctocephalites* and *Tonkinella*. The cladogram consisting of just the spinose oryctocephalids and *Tonkinella* (Fig. 2B) suggests such a relationship, but *T. spinosa* is not basal to the *Tonkinella* clade. *Tonkinella spinosa* illustrates a character state reversal from nonspinose to spinose. This is interpreted as a homeotic change where a portion of the pygidium has added thoracic spines. Homeotic changes are typically the result of mutations of HOX genes; thus, *T. spinosa* represents another example of homeotic evolution in Cambrian trilobites (see Sundberg 2000).

In both analyses, *Barklyella* falls within the *Tonkinella* clade, as suggested by Shergold (1969). However, the pygidium of *Barklyella expansa* is not known and its discovery could easily relocate the species to another part of the cladogram. The pygidium assigned to *Barklyella* sp. by Yuan *et al.* (2002, pl. 28, fig. 5) is not based on articulated or abundant specimens. This pygidium is similar to specimens of *Ovatoryctocara granulata* Chernysheva, 1962 (illustrated by Yuan *et al.* 2002, pl. 31, figs 11, 13) from the same locality (TYA14F). *Barklyella* was not included in the third cladistic analysis of the *Tonkinella* species because of the lack of a known pygidium.

Additional results from this analysis suggest that *Oryctocephaloides* is polyphyletic, with the two species being separated on the cladogram by the three *Oryctocephalites* species. The hypothesis that *Cheiruroides* gave rise to *Arthricocephalus* (Yuan *et al*. 2002) is not supported by the cladogram, as the two clades are separated by *Oryctocephaloides* and *Oryctocephalites*.

The reweighting of the characters by using the retention coefficient (RC) decreases the importance of characters that have high homoplasy. However, the reduction of the weights does not indicate that these characters were not important in defining clades (see Sundberg 2004). In the analyses, reweighting of characters had the same effect (Table 1). Those characters that maintained

Character	Type	Status	Original Weight	Final Weight	Percent of original weight
1	Ord	-	500	22	4
2	--	-	1000	1000	100
3	Ord	-	333	0	0
4	Ord	-	333	1000	300
5	Ord	-	333	1000	300
6	Ord	-	500	0	0
7	Ord	-	333	143	43
8	Ord	-	333	0	0
9	--	-	1000	0	0
10	Ord	-	500	444	89
11	--	-	1000	1000	100
12	Ord	-	1000	1000	60
13	Ord	U	1000	0	0
14	--	-	1000	1000	100
15	Ord	-	1000	476	48
16	Ord	-	333	143	43
17	Ord	-	500	0	0
18	Ord	-	333	143	43
19	Ord	-	1000	198	20

Table 3. Percentage of change in character weights resulting by progressive reweighting of characters based on RC of the *Tonkinella* cladogram. Ord = ordered character states, U = uninformative.

100 percent of their weight were: presence of the posterior lobe of the hypostome [10], triangular termination of the rostral plate [11], and pygidial convexity [19]. Most of these characters help define larger clades. The remaining characters defining larger clades had their weights reduced by 50 percent or more. Those characters that had weights reduced to 10 percent or less include: the shape of the lateral glabellar furrows [4]; direction of the ocular ridges [6]; the presence of genal spines [9] (reduced to zero); and length of the pygidial axis [18]. Most of these characters define terminal nodes (species). Of interest is the reduction to zero of character 9, which is the presence or absence of genal spines. Two clades are in part defined by the absence of genal spines, the *Cheiruroides* and *Arthricocephalus* clades. These two clades do not occur next to each other suggesting that other characters are more important in defining larger clades. A similar pattern of change in weights was observed in the analysis that included only the spinose oryctocephalids and *Tonkinella*. Four characters were uniform throughout the clade [1, 9, 10, 11] (Table 2). Those informative characters that maintained 100 percent of their weight were the presence of pygidial interpleural furrows [16] and pygidial convexity [19]. Those characters that had weights reduced to 10 percent or less include: the number of transglabellar furrows [2]; the shape of the lateral glabellar furrows [4]; and direction of ocular ridges [6] (reduced to zero).

Reweighting of the characters in the analysis that was limited to *Tonkinella* and *Metabalangia* was also done (Table 3). Those characters that maintained 100 percent of their weight or increased were: the convexity of the frontal lobe [2]; depth of the S2 transglabellar furrow [4]; depth of the S3 transglabellar furrow [5]; completeness of the pygidial margin [11]; curvature of the anterior border of the pygidium [12]; and the number of pygidial axial rings [19]. These characters help define mostly larger clades. Most of the remaining characters defining larger clades had their weights reduced by 50 percent or more. Those characters that had weights reduced to 10 percent or less include: the glabellar expansion [1]; strength of the S1 transglabellar furrow [3] (reduced to zero); length of the palpebral lobes [8] (reduced to zero); presence of a plectrum [9] (reduced to zero); depth of the pleural furrows on the pygidium [13] (reduced to zero); and width of the posterior lobe [17] (reduced to zero). Once again, these characters typically help define the terminal nodes.

SYSTEMATIC PALAEONTOLOGY

Class TRILOBITA Walch, 1771
Order CORYNEXOCHIDA Kobayashi, 1935
Family ORYCTOCEPHALIDAE Beecher, 1897
Subfamily ORYCTOCEPHALINAE Beecher, 1897

Tonkinella Mansuy, 1916

Type species. Tonkinella flabelliformis Mansuy, 1916.

Tonkinella spinosa new species (Fig. 5A-J)

Etymology. From Latin, *spina*, spine in reference to the spinose pygidium.

Diagnosis. Glabella expanding forward (120-135% basal glabellar width), frontal lobe moderately convex with no medial sulcus, S1 transglabellar furrow effaced to moderately developed, S2 and S3 very shallow, connected to axial furrows. Ocular ridges effaced, moderately posterolaterally directed (75°), palpebral lobe length 25-40% glabellar length. Plectrum absent, anterior border horizontal. Pygidium with spines on the anterior three or four pleural segments, no curvature of anterior border of pygidium, pleural furrows shallow to moderately deep, reaching

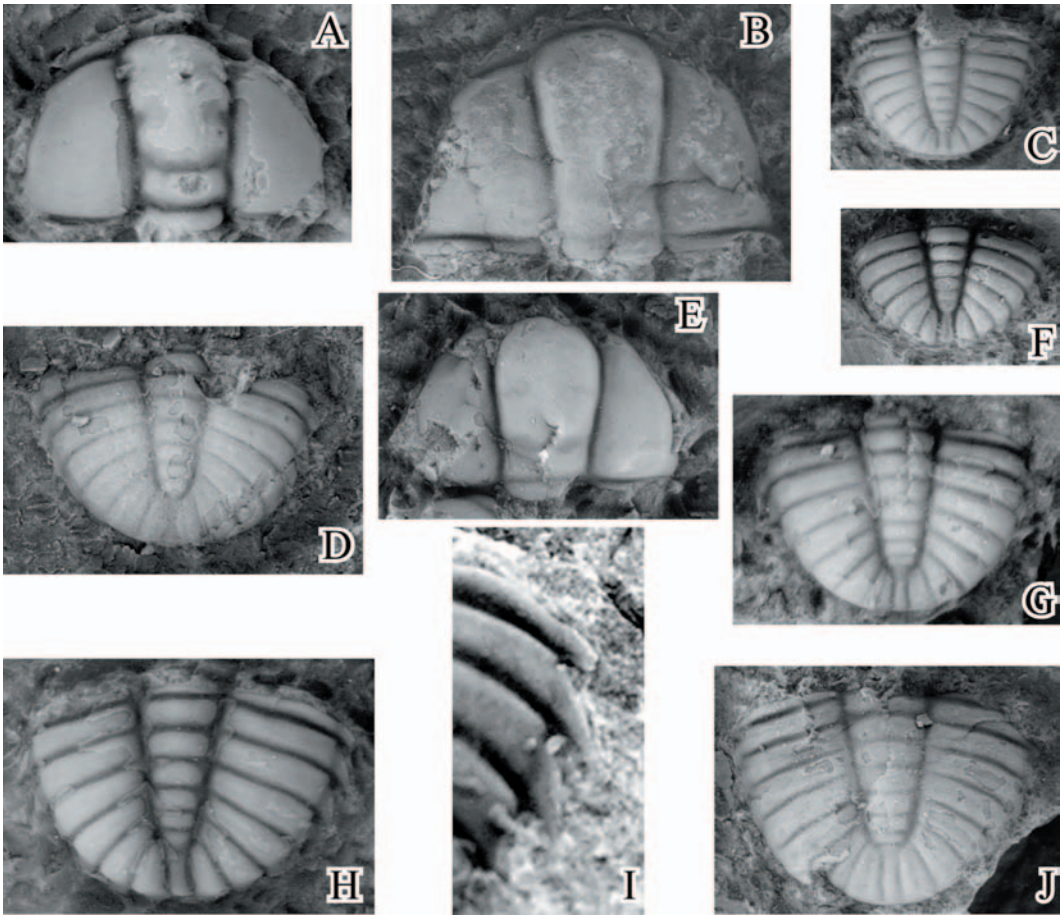

Fig. 5. Tonkinella spinosa n. sp. from the Emigrant Formation, Horse Thief Canyon, California, USA (USNM loc. 42248). Magnifications are x6.5 unless otherwise noted. A, testate cranidium with distinct S1 furrow, paratype USNM 529344. B, testate cranidium with S1 furrow absent, paratype USNM 529343. C, testate pygidium, paratype USNM 529348. D, testate pygidium, paratype USNM 529354. E, testate small cranidium, paratype USNM 529345. F, testate small pygidium, paratype USNM 529353. G, testate pygidium, paratype USNM 529347. H, I, mostly exfoliated pygidium, holotype USNM 529342, I, x 11.4. J, mostly testate pygidium, paratype USNM 529349.

margin; with 13-15 pleural ribs, anterior pleural ribs with slight backward curvature of the distal portion, posterior pleural rib moderately narrow with rib width 6 ± 1% pygidial width. Pygidial axis moderately long (75 ± 4% pygidial length), with 5-6 axial rings.

Type Material. Holotype, complete pygidium USNM 529342; paratypes USNM 529343-529354. All specimens are from USNM locality 42248.

Locality. *Glossopleura* or *Albertella* Biozone, middle carbonate member, Emigrant Formation, Horse Thief Canyon, Last Chance Range, eastern California (see below). USNM locality 42248.

Description. Cranidium small, length 2.3 to 4.8 mm (n = 4); subpentagonal; moderately convex (sag. and tr.); anterior margin moderately to strongly curved, straighter on lateral margins; posterior margin, excluding occipital ring, straight. Anterior branches of facial sutures slightly convergent to border furrow and then moderately convergent to anterior margin; posterior branches moderately divergent. Glabella elongate, moderately expanding, width at anterior end 120 to 135% glabellar width (K2); moderate convexity (sag. and tr.); frontal lobe bluntly rounded, without medial sulcus; length 95 to 96% cranidial length; basal width 25 to 35% cranidial width. Axial furrow moderately deep, slightly concave in outline; preglabellar furrow moderately shallow. Glabellar furrows moderately deep to absent, S1 directed laterally, connected with transglabellar furrow, S2 directed

slightly posteriorly, S3 and S4 poorly defined or absent. Occipital ring slightly convex, length 10 to 15% glabellar length; without occipital spine; posterior margin slightly convex. SO straight and shallow to deep. Anterior border flat, narrow, horizontal. Anterior border furrow shallow, narrow medially, wider laterally. Fixigena slightly concave, slightly downsloping, width 80 to 110% glabellar width (K2). Palpebral lobes slightly curved, horizontal, narrow and long, length 25 to 40% glabellar length; anterior end located about opposite S2 or L3. Ocular ridge effaced to weakly developed, straight, directed posterolaterally from glabella about 75º to axis. Posterior area of fixigena triangular in shape, slightly downsloping; length 15 to 30% glabellar length; width 50 to 60% glabella length.

Librigena, rostral plate, hypostome, and thorax unknown.

Pygidium moderate to small, isopygous, length 3.0 ± 1.1 mm (n = 9); semicircular, length 67 ± 5% width; moderately convex (sag. and tr.); maximum width near anterior end of axis. Marginal spines present on first to third or forth anterior pleural bands, short, broad based, slightly downsloping, pointed, directed strongly posterolaterally; posterior margin either undulatory or complete. Axis moderately tapered, anterior width 27 ± 2% pygidial width; length 75 ± 4% pygidial length; 5 to 6 axial rings, moderately convex; terminal piece small, rounded; axial ring furrows moderately deep. Pleural regions with pleural furrows moderately wide and moderately deep, extending to posterior margin, slightly curved posterolaterally; interpleural furrows absent to faint distally; anterior and posterior pleural bands not differentiated, 13-15 moderately developed pleural ribs. Terminal pleural rib is 6 ± 1% of pygidial width. Border poorly defined by flexure at the pygidial spines.

Surface of all known sclerites smooth. Exoskeleton thin.

Remarks. The spines on the pygidium of this new species are delicate and difficult to prepare on the limestone specimens. Variability in the glabellar furrows and SO is real given the good preservation of the material. Some specimens have effaced lateral glabellar furrows and transglabellar furrow and a shallow SO (Fig. 5B), and others have well defined S1 and S2 furrows and a deep SO (Fig. 5A).

Tonkinella spinosa n. sp. can be distinguished from other species of *Tonkinella* by its possession of pygidial spines on the first three or four pygidial segments. The cranidium differs from those of *T. breviceps*, *T. flabelliformis* and *T. sequei* by a more rapidly expanding glabella, and differs from *T. sibirica* by more laterally directed ocular ridges. The pygidium of *T. spinosa* is most similar to *T. sibirica* in having narrow posterior pleural segments, a long axis, and a similar number of axial rings, but it differs in having pleural furrows that reach the margin and in having 15 pleural ribs that have a slight backwards curvature. The third cladistic analysis identifies *T. stephensis* as a sister taxon to *T. spinosa*; both species share a scalloped pygidial margin. *Tonkinella spinosa* differs from this species by its pygidial spines, effaced ocular ridges, longer pygidial axis, absence of a plectrum, slight curvature of the anterior margin of the pygidium, and a narrower posterior rib of the pygidium.

Stratigraphy. The Emigrant Formation is a thick package of deep water carbonates and siliciclastics that were deposited in an outer-shelf to slope setting. The formation includes megabreccia-carbonate debris flows and pelagic and hemipelagic sediments (Albers & Stewart 1972; Kepper 1981; McCollum & Sundberg 2000). The formation ranges in age from Cambrian Series 2 (Dyeran Stage as used in Laurentia, uppermost *Olenellus* Biozone) to Lower Ordovician (Sundberg & McCollum 1997, 2003; J. F. Miller, personal communication to L. McCollum about conodonts from the upper horizons). In Horse Thief Canyon, the Emigrant Formation is approximately 700 m thick (Ansell 1987). Unfortunately, the precise horizon where specimens of *T. spinosa* were collected was not recorded by Ansell in his thesis. Trilobites co-occuring with *T. spinosa* include *Oryctocephalites walcotti* (Resser, 1938a), *Kootenia*, *Poliella*, *Pagetia* and *Syspacephalus*. Based on the associated taxa, the material came from 250 to 300 m above the base of the formation and is either from the *Albertella* or *Glossopleura* Biozone (see Sundberg & McCollum 1997, fig. 5).

CONCLUSIONS

Tonkinella was probably derived from the Oryctocephalinae as suggested by McNamara (1986) and Whittington (1995) and should be included within the subfamily. *Metabalangia* appears to be the sister group to *Tonkinella*, such a close relationship was suggested by McNamara *et al.* (2006). The phylogenetic analysis does not support having *Tonkinella* derived from the nonspinose oryctocephalids (Oryctocarinae) as suggested by Zhang (1980) or Yuan *et al.* (2001b). *Barklyella expansa* falls within the *Tonkinella* clade as suggested by Shergold (1969).

Tonkinella spinosa was initially thought to be a link between the spinose oryctocephalids and *Tonkinella*, but the pygidial spines are probably

the result of character reversal, representing a homeotic mutation.

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APPENDIX 1

Characters for the spinose and nonspinose oryctocephalid analyses. Linear measurements were made on actual specimens using an eyepiece scale with a binocular microscope or from photographs using calipers. Angles were measured using a protractor and degrees of arc were measured by overlying photographed specimens with a transparency marked with different degrees of arc. See Sundberg & McCollum (1997) for measurements.

1. Facial suture: 0 = opisthoparian, 1 = gonatoparian, 2 = proparian.
2. Number of transglabellar furrows: 1 = 1 furrow, 2 = 2 furrows, 3 = 3 furrows.
3. Number of glabellar furrows connected to axial furrows: 0 = not connected, 1 = S1 connected, 2 = S2 connected, 3 = S3 connected. Most of these furrows are connected sequentially, e.g. if the S2 is connected, then S1 is typically connected.
4. Lateral glabellar furrow slits: 0 = pits, 1 = slits.
5. Glabellar expansion: 0 = not expanded, 1 = medially expanded, 2 = anteriorly expanded.
6. Length of palpebral lobes, measured relative (%) to glabellar length: 0 = 23-31, 1 = 32-40, 2 = 41-48, 3 = 49-58.
7. Direction of ocular ridges, measured as degrees from the glabellar sagittal line: 0 = 52-60, 1 = 61-67, 2 = 68-74, 3 = 75-81.
8. Strength of ocular ridges: 0 = weak or absent (e.g., *T. breviceps*), 1 = strong (e.g., *B. expansa*).
9. Genal spines: 0 = absent, 1 = present.
10. Posterior lobe of hypostome: 0 = present, 1 = absent.
11. Termination of rostral plate triangular: 0 = absent, 1 = present.
12. Number of thoracic segments: 0 = 17-19, 1 =14-15, 2 = 12, 3 = 7-9, 4 = 5-6.
13. Pleural spines: 0 = present, 1 = absent.
14. Size of the pygidium relative to the size of the cranidium: 0 = micropygous, 1 = subisopygous, 2 = isopygous.
15. Pygidial spines: 0 = absent, 1 = present.
16. Pygidial interpleural furrows: 0 = absent, 1= present.
17. Pygidial axial segments: 1 = 1 segment, 2 = 2

segments, 3 = 3 segments, etc.

18. Length of pygidial axis, measured as axial length relative (%) to pygidial length: 0 = 40-50, 1 = 51-60, 2 = 61-70, 3 = 71-80, 4 = 81-90.

19. Pygidial convexity: 0 = low, 1 = moderate.

20. Pygidial border: 0 = absent, 1 = present.

APPENDIX 2

Character states for the taxa used in the phylogenetic analysis of the spinose and nonspinose oryctocephalids. A = *Arthricocephalus*, *Ar* = '*Arthricocephalites*', *B* = *Barklyella*, *C* = *Cheiruroides*, *Ch* = *Changaspis*, *L* = *Lancastria*, *M* = *Metabalangia*, *O* = *Oryctocephalus*, *Od* = *Oryctocephaloides*, *Ot* = *Oryctocephalites*, *T* = *Tonkinella*.

species	Characters																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
T. breviceps	0	1	3	1	2	{01}	23	0	1	0	0	5	1	2	0	0	{3456}	2	1	0
T. flabelliformis	0	1	3	1	2	0	2	0	?	0	0	?	?	2	0	0	{34}	1	1	1
T . ex. gr. *flabelliformis*	?	?	?	?	?	?	?	?	?	0	0	?	1	2	0	0	4	1	1	1
T. gavrilovae	0	3	1	1	1	{12}	0	0	?	?	?	5	0	2	0	1	6	{23}	1	0
T. sequei	0	1	?	?	2	2	?	0	?	0	0	?	?	2	0	0	3	0	1	?
T. sibirica	0	1	1	1	2	0	0	0	1	0	0	4	1	2	0	0	{67}	{23}	1	?
T. spinosa	0	1	3	1	2	{01}	?	0	?	0	0	?	?	2	1	0	{57}	3	1	0
T. stephensis	0	1	3	1	2	1	{23}	0	1	0	0	4	1	2	0	0	5	2	1	0
T. valida	0	1	3	1	2	{123}	{12}	1	?	0	0	?	?	2	0	0	{56}	2	1	1
B. expansa	0	1	3	0	2	2	{01}	0	?	?	?	?	?	?	?	?	?	?	?	?
Od. convexus	0	1	1	1	1	1	2	0	1	?	?	3	1	2	1	1	3	{34}	0	0
Od. robustus	0	1	0	1	1	{12}	0	0	1	?	?	3	1	0	1	1	2	{12}	0	0
O. lancastrioides	0	0	0	0	0	{12}	{12}	1	1	0	0	0	1	0	1	1	2	1	0	0
Ot . *rasettii*	0	{01}	{12}	{01}	2	{01}	{01}	0	1	0	0	3	1	1	1	1	2	{01}	0	0
Ot . *palmeri*	0	1	0	0	1	1	0	0	1	0	0	3	1	1	1	1	3	{23}	0	0
Ot. typicalis	0	1	0	{01}	2	{012}	{01}	0	?	0	0	?	?	1	1	1	4	{34}	0	0
M. yupingensis	0	1	0	1	1	{23}	0	1	1	?	?	4	1	2	0	1	4	{23}	0	1
Ch. placenta	0	1	0	0	2	{23}	{012}	1	1	0	0	1	1	0	1	1	1	{12}	0	0
L. roddyi	0	1	0	0	0	0	2	1	1	0	0	0	1	0	1	1	2	{23}	0	0
A. chauveaui	1	3	3	0	2	0	{23}	0	0	?	?	{43}	0	2	0	1	5	{23}	0	1
Ar. xinzhaiheensis	2	3	0	1	2	{12}	0	0	?	?	?	3	0	2	0	1	4	2	0	1
C. primigenius	1	1	3	1	0	0	{12}	0	?	?	?	2	0	0	0	1	2	?	0	0
C. arcticus	1	1	3	1	1	{01}	1	0	0	1	1	{12}	0	0	0	1	3	?	0	0
Redlichia	0	{12}	3	1	0	3	0	1	1	0	0	{01}	1	0	1	1	{12}	3	0	0

APPENDIX 3

Character state change at the nodes in Figure 1. Taxa not listed are the result of no character change from the preceding node. See Appendix 2 for generic abbreviations.

Node/Terminus	State Change	Node/Terminus	State Change
25	4(0)	46	20(0)
26	17(3)	*A. chauveaui*	3(3), 4(0), 6(0), 7(1), 17(5)
27	18(1)	*Ar. xinzhaiheensis*	1(2)
28	6(1), 17(2)	*C. arcticus*	5(1), 17(3)
29	3(0), 4(0), 6(2), 7(2), 18(2)	*Ch. elongata*	5(2), 12(1)
31	1(1), 6(0), 9(0), 10(1), 11(1), 13(0), 15(0)	*L. roddyi*	6(0)
32	6(2), 7(0), 8(0), 12(2), 17(2)	*M. yupingensis*	6(2), 8(1), 12(4)
33	12(3)	*O. lancastrioides*	2(0), 18(1)
34	6(1), 14(1), 18(2)	*Od. convexus*	3(1), 7(1)
35	7(0), 17(3)	*Od. robustus*	3(0), 5(0), 7(0)
36	18(3)	*Ot. palmeri*	4(0)
37	14(2)	*Ot. rasettii*	3(2), 5(2), 18(1)
38	1(1), 2(3), 5(2), 9(0), 13(0), 18(2)	*Ot. typicalis*	5(2), 17(4),
39	15(0), 17(4), 20(1)	*T. breviceps*	3(3), 7(2), 12(5)
41	3(3), 5(2), 16(0), 19(1)	*T. flabelliformis*	6(0), 7(2)
42	17(5)	*T. gavrilovae*	2(3), 5(2), 12(5), 13(0), 16(1)
43	7(2)	*T. sibirica*	6(0)
44	12(4)	*T. spinosa*	15(1), 18(3)
45	3(1), 17(6)	*T. valida*	8(1)

APPENDIX 4

Cladistic characters for *Tonkinella* species cladogram. Measurements are as described in Appendix 1.

1. Glabellar expansion measured as the percentage of frontal lobe width of basal glabellar width: 0 = 112-122, 1 = 123-129, 2 = 130-136.
2. Frontal lobe convexity: 0 = low, 1 = moderate.
3. Depth of the S1 transglabellar furrow: 0 = effaced, 1 = shallow, 2 = moderate, 3 = deep.
4. Depth of the S2 transglabellar furrow: 0 = effaced, 1 = shallow, 2 = moderate, 3 = deep.
5. Depth of the S3 transglabellar furrow: 0 = effaced, 1 = shallow, 2 = moderate, 3 = deep.
6. Strength of ocular ridges: 0 = effaced, 1 = weak, 2 = moderate.
7. Direction of ocular ridges, measured as degrees from the glabellar sagittal line: 0 = 52-61, 1 = 62-68, 2 = 69-75.
8. Length of palpebral lobes, measured relative (%) to glabellar length: 0 = 27-36, 1 = 37-45, 2 = 46-53.
9. Plectrum: 0 = absent, 1 = present.
10. Anterior border slope: 0 = horizontal, 1 = slight dorsal tilt, 2 = upturned.
11. Pygidial margin: 0 = smooth (complete), 1 = scalloped.
12. Curvature of the anterior border of the pygidium: 0 = no curvature, 1 =slight curvature, 2 = strong curvature.
13. Depth of the pygidial pleural furrows: 0 = moderately shallow, 1 = moderate depth.
14. Pleural furrows reaching the pygidial margin: 0 = no, 1 = yes.
15. Pleural rib count of the pygidium: 0 = 9, 1 = 11, 2 = 12, 3 = 13, 4 = 14, 5 = 15.
16. Curvature of the lateral pleural ribs: 0 = none (e.g., *T. stephensis*), 1= slight, 2 = moderate, 3 = strong (e.g., *T. flabelliformis*).
17. Width of the posterior pleural rib measured as lobe width relative (%) to pygidial width: 0 = 27-36, 1 = 37-45, 2 = 46-54.
18. Length of pygidial axis, measured as axial length relative (%) to pygidial length: 0 = 40-49, 1 = 50-59, 2 = 60-69, 3 = 70-79.
19. Number of pygidial axial rings: 2 = 2 rings, 3 = 3 rings, 4 = 4 rings, 5= 5 rings, 6 = 6 rings.

APPENDIX 5

Character states for the taxa used in the phylogenetic analysis. *M* = *Metabalangia*, and *T* = *Tonkinella*.

	Characters																		
species	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
T. breviceps	{01}	1	2	1	1	1	{23}	{01}	0	2	0	0	{01}	0	{123}	2	0	{23}	{345}
T. flabelliformis	0	1	2	1	1	1	2	0	0	?	0	2	1	0	1	3	{01}	1	4
T . ex. gr. *flabelliformis*	?	?	?	?	?	?	?	?	?	?	0	1	1	0	1	2	0	1	4
T. gavrilovae	{01}	0	3	3	{23}	1	0	{01}	0	2	0	0	1	0	3	0	0	{23}	6
T. sequei	0	1	1	1	1	1	?	2	0	?	0	?	0	?	0	0	2	0	2
T. sibirica	2	1	2	2	2	1	0	0	0	?	0	0	1	0	3	0	0	{23}	6
T. spinosa	{12}	1	{012}	1	1	0	2	{01}	0	0	1	0	{01}	1	{345}	1	0	3	{56}
T. stephensis	{01}	1	2	1	1	1	{23}	1	1	0	1	0	1	1	3	0	1	2	{45}
T. valida	{012}	0	2	2	2	2	{012}	{123}	1	1	0	0	{01}	0	3	{01}	2	2	{56}
M. yupingensis	{012}	1	3	0	0	2	1	{12}	0	0	0	1	2	1	?	1	?	{2,3}	4

APPENDIX 6

Character state change at the nodes in Figure 4. Taxa not listed is the result of no character change from the proceeding node. See Appendix 5 for generic abbreviations.

Node/Terminus	State Change	Node/Terminus	State Change
11	8(0), 15(1), 16(2)	*T. breviceps*	10(2), 16(2)
12	5(1), 6(1), 14(0), 18(1)	*T. flabelliformis*	12(2), 16(3)
13	3(2), 4(1), 7(2), 13(1)	*T. gavrilovae*	3(3), 4(3), 10(2)
14	11(1)	*T. sequei*	3(1), 8(2), 13(0), 16(0), 17(2), 18(0), 19(2)
15	1(1), 12(0), 15(3), 19(5)	*T. sibirica*	1(2), 8(0)
16	2(0)	*T. spinosa*	6(0), 18(3)
17	4(2), 5(2), 7(0), 10(1), 16(0), 19(6)	*T. stephensis*	9(1), 16(0), 17(1)
18	5(1), 6(1), 14(0)	*T. valida*	6(2), 9(1), 17(2)

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